

Retail Customer Analytics: From Data to Business Insights

1. Project Overview

This project analyzes customer shopping behavior using a transactional dataset containing 3,900 customer purchase records. The purpose of the project is to turn raw shopping data into clear business insights about revenue, customer behavior, product performance, discounts, loyalty, subscriptions, shipping preferences, and age-based purchasing patterns. The project was designed as a complete analytics workflow rather than a single-tool exercise. Python and pandas were used to inspect, clean, and engineer the data; the cleaned DataFrame was transferred into Microsoft SQL Server for structured analysis; and Power BI was connected to the SQL Server table to build an interactive dashboard for business-facing interpretation.

A key objective was to keep each stage connected. Instead of performing cleaning separately and manually importing a different version of the file into every tool, the cleaned Python DataFrame was written directly to the SQL Server database. This created a simple data pipeline in which the same prepared data could be queried in SQL Server Management Studio (SSMS) and visualized in Power BI. The final dashboard therefore reflects the cleaned and engineered dataset used in the SQL analysis.

2. Dataset Summary

The original dataset contains 3,900 rows and 18 columns. Each row represents one customer purchase record and includes demographic information, product details, payment and shipping information, review ratings, subscription status, discount usage, previous purchase history, and frequency of purchases.

Dataset Characteristics	Value
Rows	3.900
Original columns	18
Columns after feature engineering and removal of redundant 'promo code used' column	19
Product category	4(Clothing, Accessories, Footwear, Outwear)
Unique products	25
Missing values identified before cleaning	37 values in Review Rating
Total revenue in dataset	\$233,081
Average purchase amount	\$59.76
Average review rating after cleaning	3.75/5

The key features can be grouped into three broad areas: customer demographics (age, gender, location, and subscription status), purchase information (item purchased, category, purchase amount, size, color, season, and payment method), and shopping behavior (discount applied, promo code used, previous purchases, frequency of purchases, review rating, and shipping type). This combination makes the dataset useful for both descriptive reporting and customer-behavior analysis.

3. Exploratory Data Analysis using Python

The Python stage focused on understanding the raw file, fixing data-quality issues, creating analysis-friendly features, and preparing a consistent table for SQL Server. The work was completed with pandas, while SQLAlchemy and pyodbc were used later to transfer the cleaned DataFrame into SQL Server.

- **Data Loading:** The CSV file was loaded into a pandas DataFrame with `pandas.read_csv()`. This created the working object `raw_data`, which was used throughout the cleaning process.
- **Initial Exploration:** `head()`, `info()`, `describe(include = "all")`, and `isnull().sum()` were used to understand the first records, data types, distributions, categorical values, and missing data. This step confirmed that the dataset contained 3,900 records and identified 37 missing Review Rating values.
- **Missing Data Handling:** Missing Review Rating values were filled with the median review rating from the corresponding product category. A category-level median was used instead of a single overall value so that the replacement remained more representative of the type of product being reviewed.
- **Column Standardization:** All column names were converted to lowercase, spaces were replaced with underscores, and `purchase_amount_(usd)` was simplified to `purchase_amount`. This made the field names easier and safer to use in SQL queries and Power BI.

```
hello.py > ...
1 import pandas
2 """ Load the customer shopping behavior dataset into pandas DataFrame"""
3 raw_data = pandas.read_csv('customer_shopping_behavior.csv')
4 """ Configure pandas to display all columns when printing the DataFrame"""
5 pandas.set_option("display.max_columns",None)
6 """ Preview the first five rows to understand the structure of the dataset"""
7 print(raw_data.head())
8 """ Display column names, data types, non-null counts, and memory usage"""
9 print(raw_data.info())
10 """ Generate descriptive statistics for both numerical and categorical columns"""
11 print(raw_data.describe(include = 'all'))
12 """ Check the number of missing values in each column before data cleaning"""
13 print(raw_data.isnull().sum())
14 """ Fill missing Review Rating values using the median rating
15 Calculated within each product category """
16 raw_data['Review Rating'] = raw_data.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
17 """ Verify that missing Review Rating values were successfully handled"""
18 print(raw_data.isnull().sum())
19 """ Standardize column names by converting them to lowercase and replacing spaces with underscores for consistency and ease of use"""
20 raw_data.columns = raw_data.columns.str.lower()
21 raw_data.columns = raw_data.columns.str.replace(' ', '_')
22 """ Display the updated column names to confirm the changes"""
23 print(raw_data.columns)
24 """ Rename the column 'purchase_amount_(usd)' to 'purchase_amount' for clarity and consistency"""
25 raw_data = raw_data.rename(columns = {'purchase_amount_(usd)': 'purchase_amount'})
26 """ Confirm that the column was renamed successfully"""
27 print(raw_data.columns)
```

- **Age Group Feature Engineering:** pandas.qcut() divided customers into four approximately equal-sized age groups. The resulting ranges were Young Adult (18-31), Adult(32-44), Middle Aged (45-57), and Senior (58-70). The final counts were 1,028 Young Adults, 942 Adults, 986 Middle Aged customers, and 944 Seniors.
- **Purchase Frequency Feature Engineering:** Text labels such as Weekly, Fortnightly, Monthly, Quarterly, Bi-Weekly, Annually, and Every 3 Months were mapped to approximate numbers of days (7,14,30,90,14,365, and 90 respectively). This created a numerical purchase_frequency_days field that is easier to use in analysis.
- **Redundancy Check:** discount_applied and promo_code used contained duplicate information. After confirming that the fields represented the same yes/no behavior, promo_code_used was removed so the final dataset would not carry an unnecessary duplicate column.
- **Validation:** After cleaning and feature engineering, missing values were checked again. The final working DataFrame had no remaining null values, including purchase_frequency_days.

```

28 """ Create four age groups using quartile-based segmentation.
29 - qcut divides customers into four groups containing roughly equal numbers of customers based on their age. """
30 labels = ['Young Adult', 'Adult', 'Middle Aged', 'Senior']
31 raw_data['age_group'] = pandas.qcut(raw_data['age'], q = 4, labels = labels)
32 """ Preview the relationship between customer age and the newly created age group. """
33 print(raw_data[['age', 'age_group']].head(10))
34 """ Convert text-based purchase frequency values into approximate numbers of days.
35 This creates a numerical feature that can be used more easily in analysis. """
36 frequency_mapping = {
37     'Fortnightly': 14,
38     'Weekly': 7,
39     'Monthly': 30,
40     'Quarterly': 90,
41     'Bi-Weekly': 14,
42     'Annually': 365,
43     'Every 3 Months': 90
44 }
45 raw_data['purchase_frequency_days'] = raw_data['frequency_of_purchases'].map(frequency_mapping)
46 """ Verify that purchase frequency categories were converted correctly """
47 print(raw_data[['frequency_of_purchases', 'purchase_frequency_days']].head(10))
48 """ Compare discount_applied and promo_code used before removing redundant data. """
49 print(raw_data[['discount_applied', 'promo_code_used']].head(10))
50 """ Remove promo_code_used because it contains duplicate information
51 already represented by the discount_applied column. """
52 raw_data = raw_data.drop('promo_code_used', axis = 1)
53 """ Display the final set of columns after data cleaning and feature engineering to confirm the changes. """
54 print(raw_data.columns)

```

- **Database Integration:** SQLAlchemy and pyodbc were used with ODBC Driver 18 for SQL Server. The cleaned DataFrame was written into the CustomerShoppingDB database as the customer_shopping_behavior table. A trusted Windows connection was used, and if_exists = “replace” ensured that rerunning the Python pipeline replaced the SQL table with the newest cleaned version.

The Python stage therefore did more than clean a CSV. It created a repeatable bridge between data preparation and database analysis. Once the script completed, SSMS could query the

same cleaned data directly, and Power BI could connect to the same SQL Server database rather than relying on a separate manual copy.

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in Microsoft SQL Server Management Studio 22 using the cleaned customer_shopping_behavior table. The queries used aggregation functions such as SUM(), AVG(), COUNT(), conditional logic with CASE, filtering with WHERE, grouping with GROUP BY, a subquery for comparison with the overall average, common table expressions (CTEs), and ROW_NUMBER() with PARTITION BY for within-category ranking. The following business questions were answered.

1. **Revenue by Gender** – SUM(purchase_amount) was calculated after grouping records by gender.

Gender	Total Revenue
Female	\$75,191
Male	\$157,890

Result and Interpretation: Male customers generated \$157,890 in revenue, and female customers generated \$75,191. Male customers therefore contributed about 67.74% of total revenue, compared with 32.26% from female customers. This difference should be interpreted together with the number of customers in each gender group; the dataset contains more male records, so the result reflects both group size and spending.

2. **High-Spending Discount Users** – A subquery first calculated the overall average purchase amount. The outer query then filtered records where a discount was applied and the purchase amount was greater than that average.

Metric	Result
Overall average purchase amount	\$59.76
Discounted purchases above average	839

Result and Interpretation: The overall average purchase amount was approximately \$59.76, and 839 discounted purchases were still above this threshold. This shows that discount use is not limited to low-value transactions; a meaningful number of discounted purchases remained higher than the typical purchase value.

3. **Top 5 Products by Rating** – Average review rating was calculated for every item; the results were sorted from highest to lowest, and TOP 5 returned the five strongest products.

Item Purchased	Average Review Rating
Gloves	3.86
Sandals	3.84
Boots	3.82

Hat	3.80
Skirt	3.78

Result and Interpretation: Gloves ranked first with an average rating of 3.86, followed by Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78). The differences are small, so the products are relatively close in customer satisfaction. This is why the Power BI chart later displays two identical places rather than relying only on bar length.

4. **Shipping Type Comparison** – The query filtered the data to Standard and Express shipping, grouped by shipping type, and compared the average purchase amount for the two groups.

Shipping Type	Average Purchase Amount
Express	\$60.48
Standard	\$58.46

Result and Interpretation: Express-shipping purchases averaged \$60.48, while Standard-shipping purchases averaged \$58.46. Express is only about \$2.02 higher, so the difference exists but is modest. It would not be enough on its own to claim that shipping choice causes higher spending.

5. **Subscribers vs Non-Subscribers** - Customers were grouped by subscription status. COUNT() measured the group size, AVG() measured average spend, and SUM() measured total revenue.

Subscription Status	Customers	Average Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Result and Interpretation: Subscribers numbered 1,053 with an average spend of \$59.49 and total revenue of \$62,645. Non-subscribers numbered 2,847, with average spend of \$59.87 and total revenue of \$170,436. Average spending is almost the same, and non-subscribers are actually slightly higher. The much larger non-subscriber revenue is mainly explained by the fact that non-subscribers represent roughly 73% of customers, while subscribers represent roughly 27%.

6. **Discount-Dependent Products** – A CASE expression converted each discounted purchase into 1 and each non-discounted purchase into 0. SUM() counted discounted transactions, COUNT(*) counted all purchases for the product, and the ratio was converted into a percentage.

Product	Total Purchases	Discounted Purchases	Discount %
Hat	154	77	50.00%

Sneakers	145	72	49.66%
Coat	161	79	49.07%
Sweater	164	79	48.17%
Pants	171	81	47.37%

Result and Interpretation: Hat had the highest discount-use rate at 50.00%, followed by Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%). These products are not necessarily unprofitable, but they appear more dependent on discounted transactions than other products and should be monitored when designing promotions.

7. **Customer Segmentation** – A CTE created a customer_segment field with CASE: 1 previous purchase labeled New, 2-10 was Returning, and more than 10 was treated as Loyal. The outer query then counted each segment.

Customer Segment	Number of Customers
Loyal	3,116
Returning	701
New	83

Result and Interpretation: The dataset contains 3,116 Loyal customers, 701 Returning customers, and 83 New customers. This is approximately 79.90% Loyal, 17.97% Returning, and 2.13% New. The strong concentration in the Loyal segment suggests that the dataset contains many established buyers. This creates an opportunity for retention- and loyalty-focused campaigns rather than relying only on acquisition.

8. **Top 3 Products per Category** – The first CTE counted purchases for each product within each category. A second CTE used ROW_NUMBER() OVER (PARTITION BY category ORDER BY total purchases DESC, item name ASC) to rank products separately inside each category. This final query kept ranks 1 through 3.

Rank	Category	Item Purchased	Purchases
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outwear	Jacket	163
2	Outwear	Coat	161

Result and Interpretation: Accessories were led by Jewelry (171), Belt (161), and Sunglasses (161); Clothing by Blouse (171), Pants(171), and Shirt(169); Footwear by Sandals (160), Shoes (150), and Sneakers (145); Outwear contained only Jacket (163) and Coat (161), so only two products could be returned for that category. This query is useful because it avoids a common mistake: finding the top products overall instead of finding the top products within each category.

- 9. Repeat Buyers & Subscriptions** – Records with more than five previous purchases were treated as repeat buyers. These records were grouped by subscription status and counted.

Subscription Status	Repeat Buyers	Share of Repeat Buyers
No	2,518	72.44%
Yes	958	27.56%

Result and Interpretation: There were 3,476 repeat buyers in total: 2,518 were not subscribed, and 958 were subscribed. That is, approximately 72.44% were not subscribed, and 27.56% were subscribed. Based on this dataset, repeat purchasing does not automatically translate into subscription adoption. This suggests that a large pool of frequent buyers may still be available for targeted subscription conversion campaigns.

- 10. Revenue by Age Group** – The engineered age_group field was grouped and purchase amounts were summed to measure each group’s contribution to revenue.

Age Group	Customers	Total Revenue
Young Adult	1,028	\$62,143
Middle Aged	986	\$59,197
Adult	942	\$55,978
Senior	944	\$55,763

Result and Interpretation: Young Adults generated \$62,143, Middle-Aged customers \$59,197, Adults \$55,978, and Seniors \$55,763. Revenue is relatively balanced across the four groups, but Young Adults contribute the largest share at about 26.66%. Because the age groups were created using quantities, these labels describe dataset-specific groups rather than universal age definitions.

5. Dashboard in Power BI

The final stage of the project was an interactive Power BI dashboard connected to the CustomerShoppingDB SQL Server database. The dashboard was designed to give a business user an immediate overview while still allowing filtering by subscription status, gender, product category, and shipping method. Because the report is based on the full cleaned SQL table rather than isolated screenshots of query results, the visuals respond interactively when slicers are selected.

Three KPI cards summarize the overall dataset: 3.9K total customers, \$59.76 average spend per purchase, and approximately \$233K total revenue. The visual layer then expands the analysis through subscription share, revenue and purchase volume by category, revenue and purchase volume by age group, and the top five highest-rated products. The dashboard uses a consistent pink/magenta design, rounded visual containers, and clear labels to keep the report approachable for non-technical viewers.



The dashboard supports several useful interpretations at a glance. Clothing is the strongest category by both revenue and the number of purchases, while Outerwear is the smallest category. The subscription donut chart shows that roughly 27% of customers are subscribed and 73% are not. Young adults generate the most revenue and also have the highest purchase count, although the four age groups remain relatively close. Finally, the highest-rated products have average ratings between 3.78 and 3.86, reinforcing the SQL finding that rating differences among the top products are small.

The Power BI stage also demonstrates how the earlier Python and SQL work becomes useful to an end user. Python created clean, analysis-ready fields such as `age_group`; SQL validated and explored business questions; and Power BI converted the same data into an interactive report where a user can explore combinations of category, gender, subscription status, and shipping method without rewriting queries.

6. Business Recommendations

- **Boost Subscription Conversion:** Only about 27% of customers are subscribers, and even among repeat buyers, only about 27.56% are subscribed. Target frequent non-subscribers with concrete benefits such as member-only discounts, free or faster shipping thresholds, early access, or loyalty points. This audience is already demonstrating purchasing behavior, so conversion may be more efficient than targeting entirely new customers.
- **Protect and Reward Loyal Customers:** Around 79.90% of records fall into the Loyal segment under the project's segmentation rules. Loyalty rewards, personalized offers, and retention campaigns can help preserve this valuable base while encouraging Returning customers to move into the Loyal group.
- **Review Discount Strategy:** Hat, Sneakers, Coat, Sweater, and Pants have the highest percentages of discounted purchases. Promotions for these products should be evaluated together with margin data before increasing discount intensity. Discounts can support volume, but repeated reliance on them can reduce profitability.
- **Prioritize High-Performing Categories:** Clothing produces the highest revenue and the largest number of purchases, followed by Accessories. Inventory planning and campaigns should ensure these categories remain well stocked while also investigating whether Footwear and Outerwear can be improved through product mix, placement, or promotion.
- **Use Product Ratings in Merchandising:** Gloves, Sandals, Boots, Hat, and Skirt are the top-rated products in the analysis. These products can be featured in recommendation sections, advertising, or cross-selling campaigns. Because the differences in rating are small, rating should be combined with sales volume and margin before making major assortment decisions.
- **Target Age Groups without Ignoring Balance:** Young Adults contribute the largest revenue amount at \$62,142 and the most purchases at 1,028. They are a string marketing target, but revenue across all four age groups is relatively close, so the business should avoid overconcentrating on only one demographic.
- **Test Shipping-Based Offers Carefully:** Express purchases have a slightly higher average purchase amount than Standard purchases (\$60.48 versus \$58.46). This may justify testing shipping incentives, but the difference is small and should not be interpreted as evidence that Express shipping itself causes higher spending.

Overall, the project shows how a simple transactional dataset can be transformed into a practical decision-support workflow. The strongest value comes from combining tools: Python improved data quality and created useful analytical features, SQL Server provided transparent and reproducible business logic, and Power BI translated the results into an interactive dashboard that can be understood without reading code. The same structure can be extended in future work by adding profit or cost data, customer-level longitudinal records, dates for time-series analysis, and predictive models for churn, subscription conversion, or purchase propensity.